Solution Architecture

Date	02 July 2026
Team ID	xxxxxx
Project Name	Smart Lender – Loan Eligibility Prediction Using Machine Learning
Maximum Marks	5 Marks

Solution Architecture Diagram:
Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.

Presentation / Client Layer
HTML5,CSS3,JavaScript
Loan Eligibility Web Interface

↓↑

API Gateway / Load Balancer
Flask Application
HTTP Request Handling
Routing

↓↑

input processing layer	**Core Logic Service**	**External API Integrations**
Input Validation *Form Validation* *User Request Handling*	*Data Preprocessing* *Feature Encoding* *Feature Scaling* *XGBoost Prediction*	*Not Used* *(Future: Banking APIs)*

↓↑

Data / Storage Layer
Loan Eligibility Dataset (CSV)
model.pkl
scaler.pkl

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface for entering applicant details, validating user input, and displaying loan eligibility prediction results through a responsive web interface.	HTML5, CSS3, JavaScript
Application Layer (Flask)	Receives HTTP requests, validates applicant information, performs data preprocessing, interacts with the machine learning model, and returns prediction results to the user.	Python, Flask
Machine Learning Service	Preprocesses applicant data, loads the trained XGBoost model and scaler, performs loan eligibility prediction, and generates Approved or Rejected results.	XGBoost, Scikit-learn, Pandas, NumPy, Joblib
Data Storage Layer	Stores the loan eligibility dataset used for training along with the trained machine learning model (model.pkl) and preprocessing scaler (scaler.pkl).	CSV Dataset, Pickle (.pkl) Files